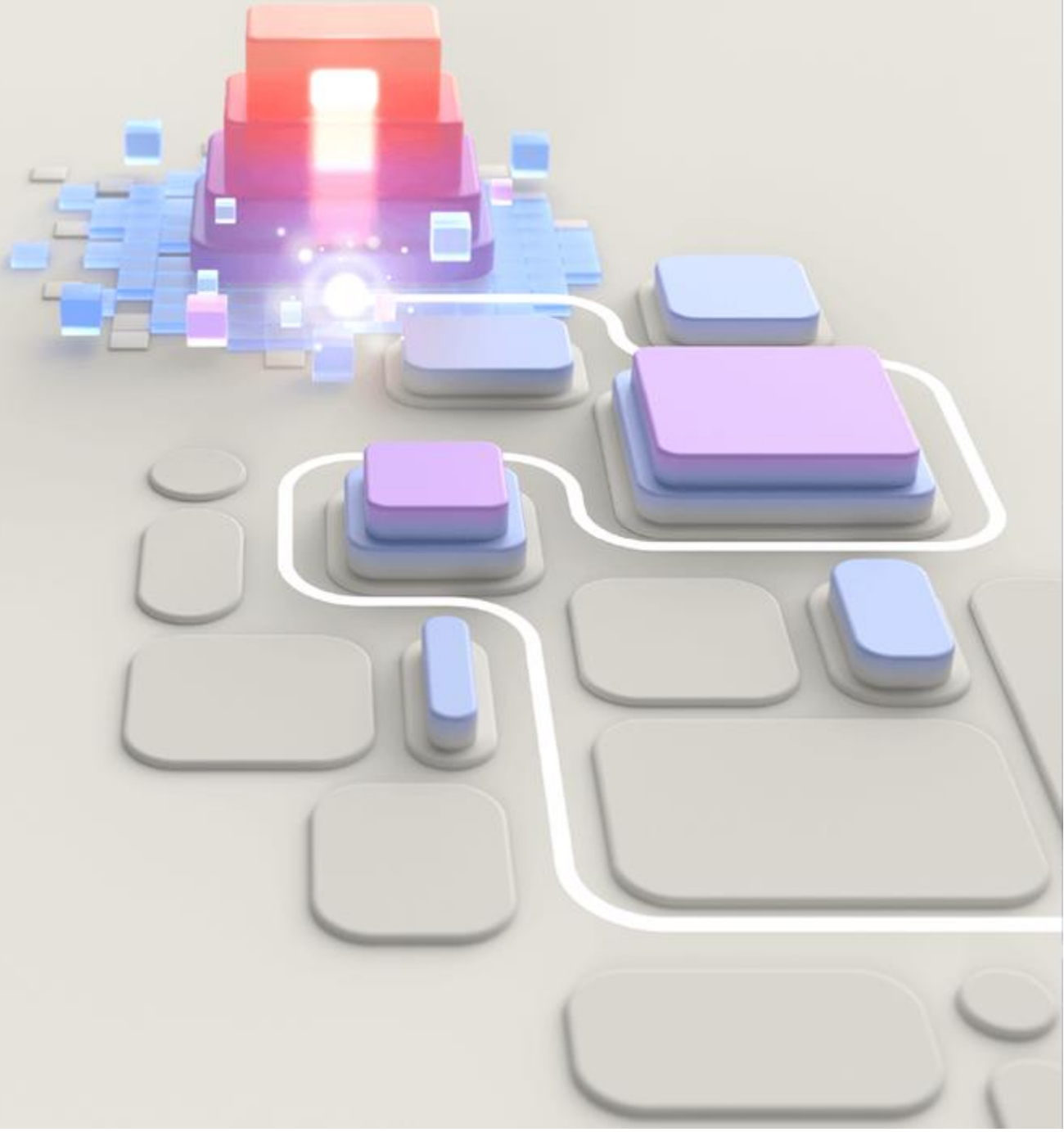
Microsoft

AZ-104

Administer Intersite Connectivity



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AZ-104 Agenda

01: Administer Identity
02: Administer Governance and Compliance
03: Administer Azure Resources
04: Administer Virtual Networking
05: Administer Intersite Connectivity
06: Administer Network Traffic Management
07: Administer Azure Storage
08: Administer Azure Virtual Machines
09: Administer PaaS Compute Options
10: Administer Data Protection
11: Administer Monitoring

Peering , Private EP

Learning Objectives – Administer Intersite Connectivity

- Configure VNet Peering
- Manage and control traffic flow with routes
- Lab 05 – Implement Intersite Connectivity

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Configure VNet Peering

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Learning Objective – Configure VNet Peering

- Determine VNet Peering Uses
- Determine Gateway Transit and Connectivity Needs
- Create VNet Peering
- Determine Service Chaining Uses
- Demonstration – Virtual Network Peering
- Learning Recap

Exam objectives

Configure and manage virtual networks in Azure

- Create and configure virtual network peering

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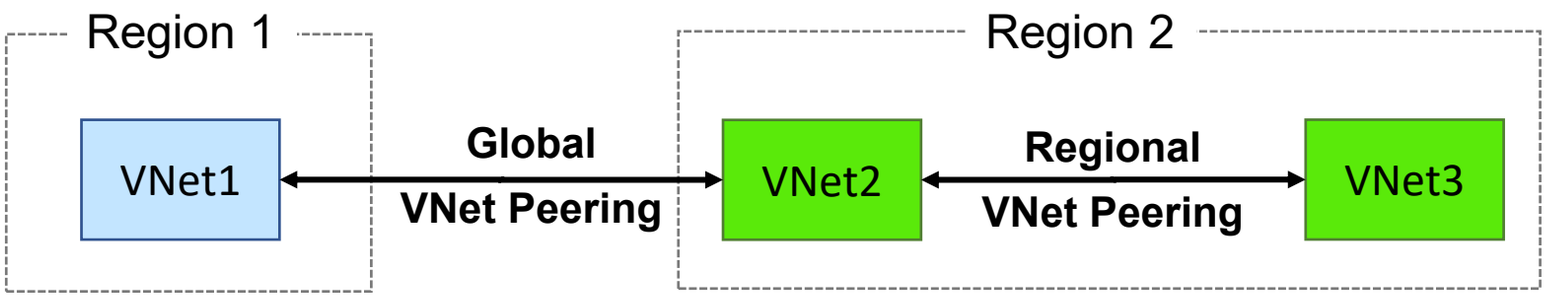
Determine VNet Peering Uses



- Two types of peering: Global and Regional
- Connects two Azure virtual networks – you can peer across subscriptions and tenants
- Peered networks use the Azure backbone for privacy and isolation
- Easy to setup, seamless data transfer, and great performance

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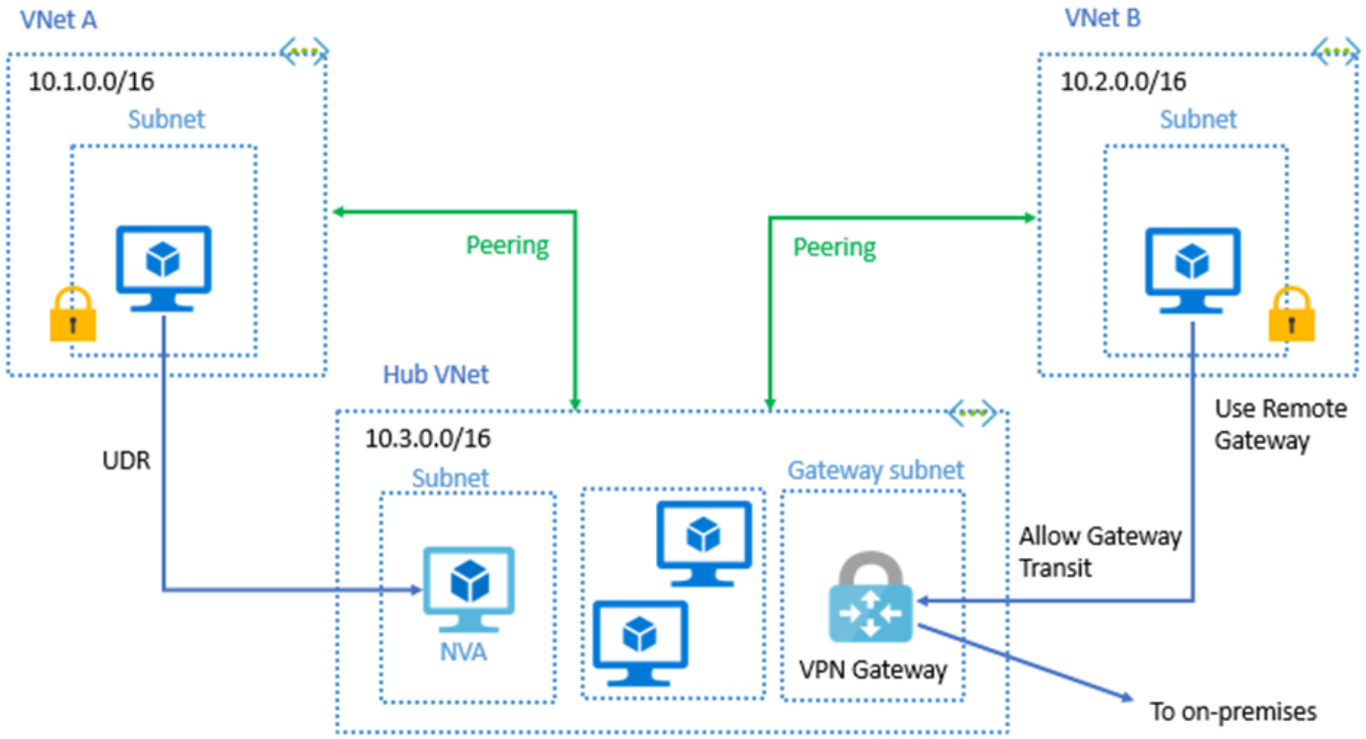
Determine Gateway Transit and Connectivity Needs

- Gateway transit allows peered virtual networks to share the gateway and get access to resources
- No VPN gateway is required in the peered spoke virtual network
- Default VNet peering provides full connectivity



IP address spaces of connected networks can't overlap

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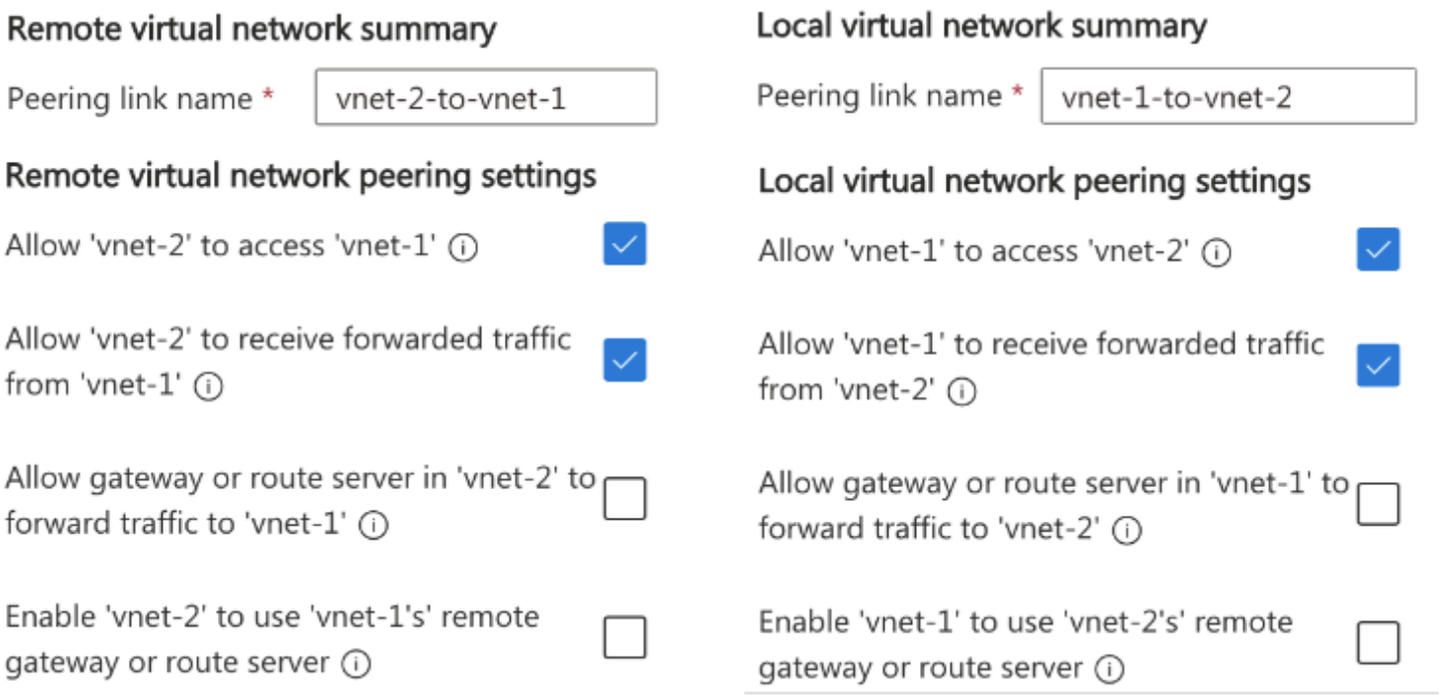
Create VNet Peering

- Allow virtual network traffic
- Allow forwarded traffic
- Allow gateway or route server traffic
- Status should show “connected”



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Manage and control traffic flow with routes

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Learning Objective - Routing and endpoints

- What are System Routes?
- What are User-Defined Routes?
- Demonstration – Custom Routing tables
- What is a Service Endpoint?
- What is Azure Private Link?
- Learning Recap

Exam objectives

Configure and manage virtual networks in Azure

- Configure user-defined network routes

Configure secure access to virtual networks

- Configure service endpoints for Azure platform as a service (PaaS)
- Configure private endpoints for Azure PaaS

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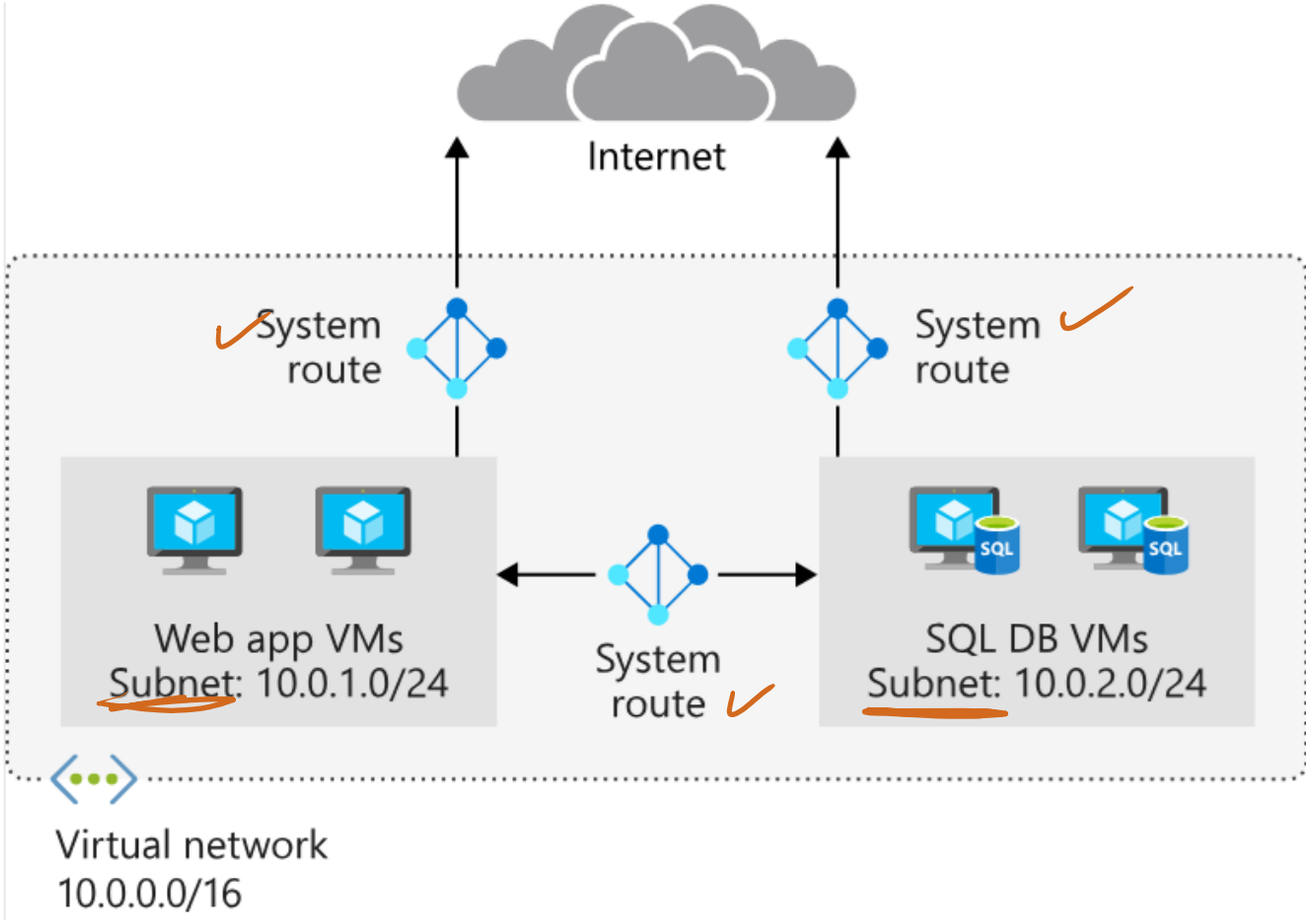
What are System Routes?

- Pre-defined routes automatically added to every subnet's route table by Azure
- Ensures fundamental connectivity (e.g., VNet-to-VNet, VNet-to-Internet, VNet-to-Azure services)
- **Examples**: Routes for VNet peering, local traffic, and the 0.0.0.0/0 internet route
- You can't delete system routes, but you *can* override with custom routes (UDRs)



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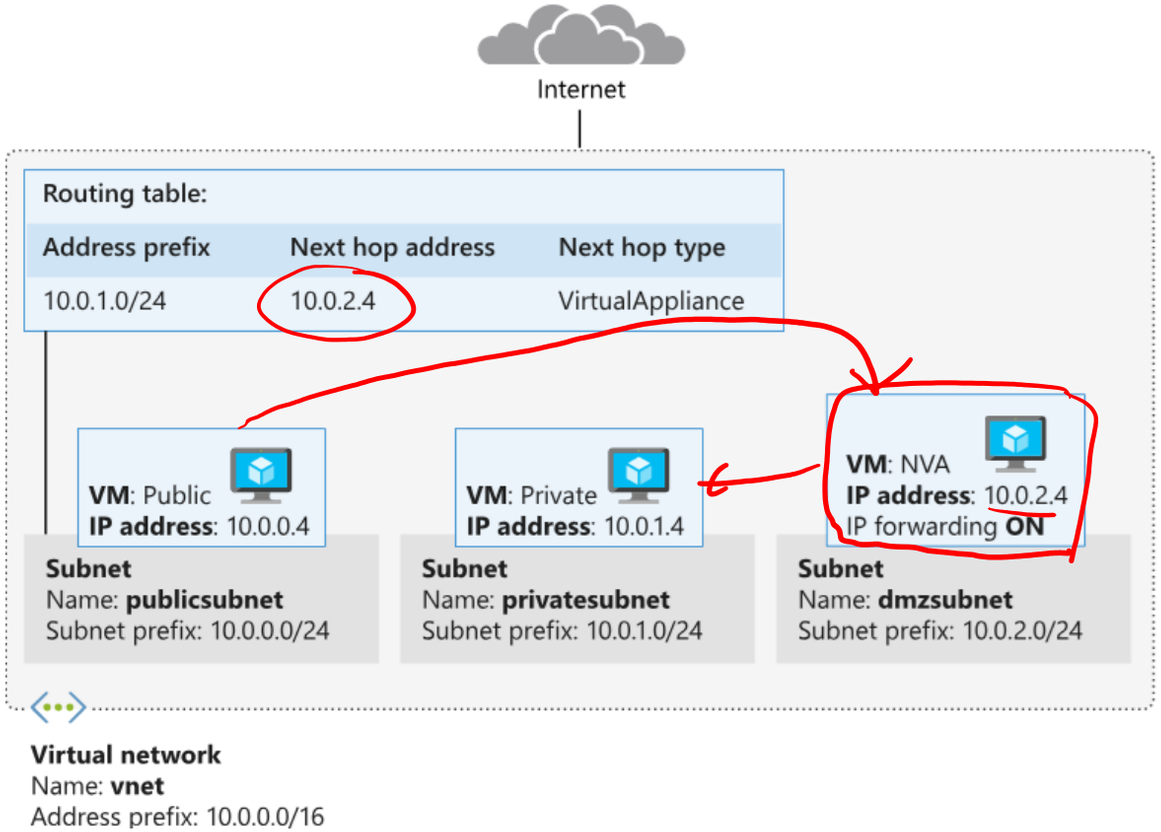
What are User-Defined Routes (UDRs)?

- UDRs override the default system routes
- Traffic can be directed to virtual appliance, Virtual Network Gateway, Virtual Network, Internet, or None (drop it)
- Border Gateway Protocol (BGP) exchanges routes between Azure and on-premises networks
- You can specify a service tag as the address prefix for a user-defined route instead of an explicit IP range



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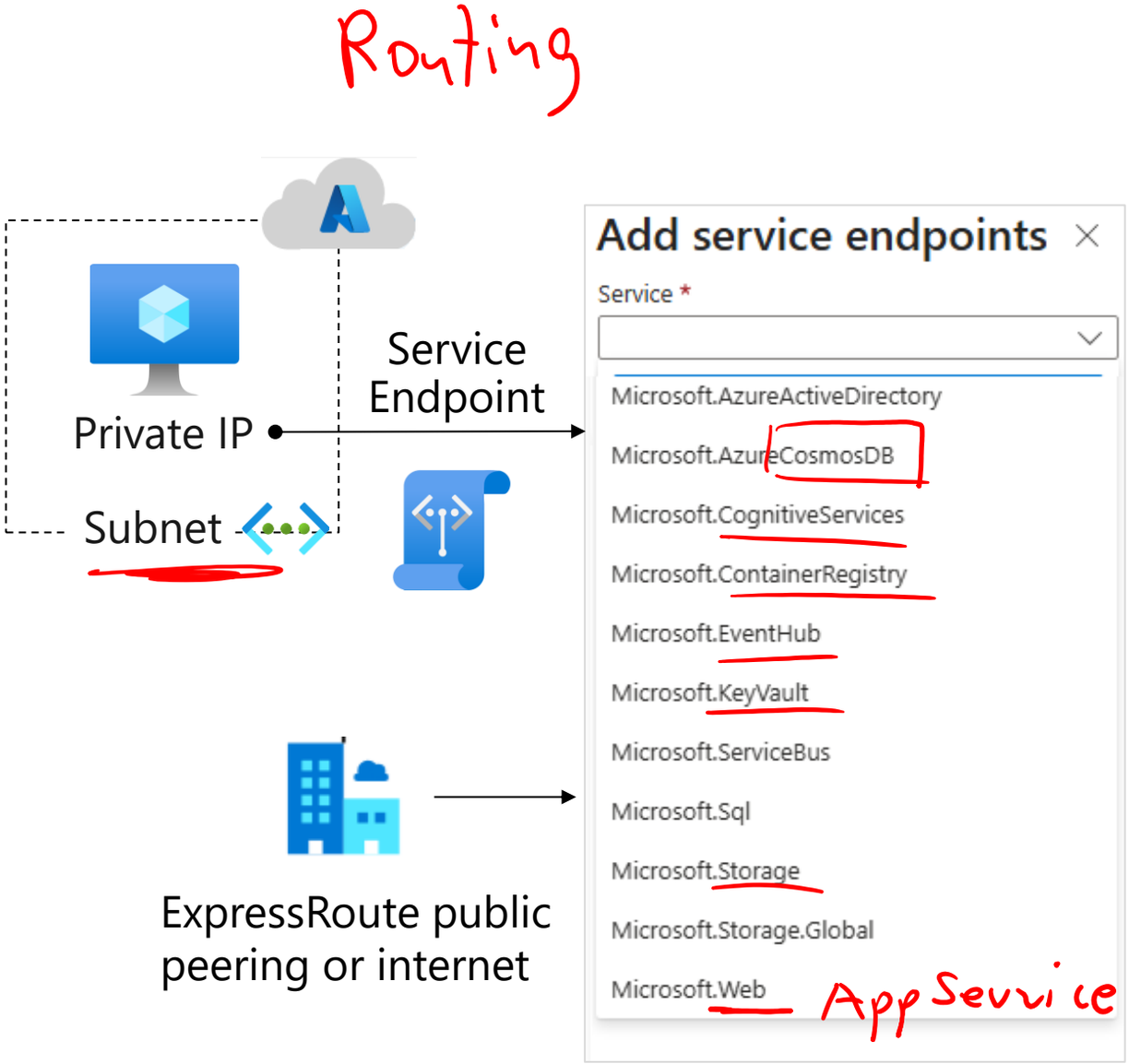
What is a Service Endpoint?

- A direct connection from your virtual network to supported Azure services
- Restricts the flow of traffic and enables your Azure VMs to access the service directly from your private address space
- Traffic always stays on the Microsoft Azure backbone network



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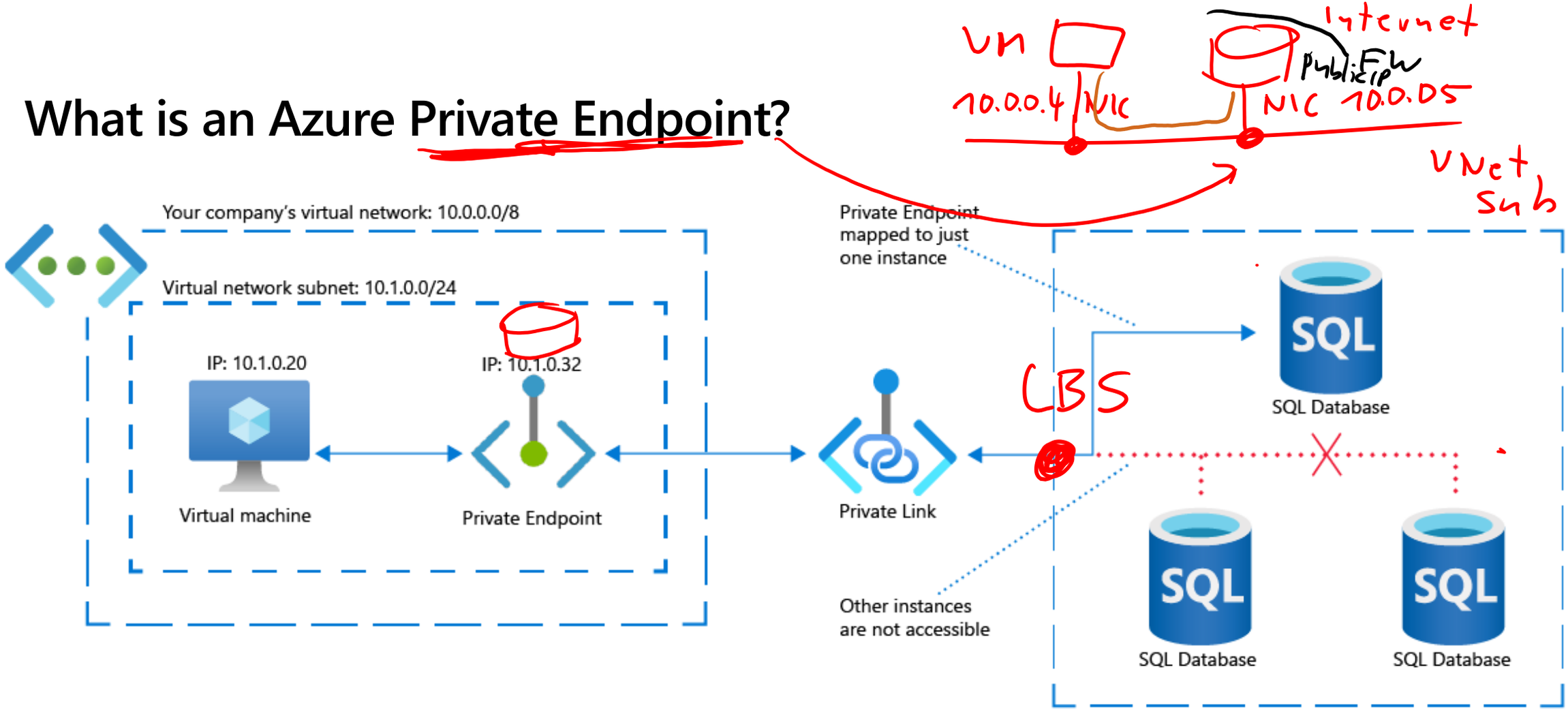
What is an Azure Private Endpoint?



Uses a private IP address from your virtual network

Granular access to services

Eliminates the potential security risk of a public IP address

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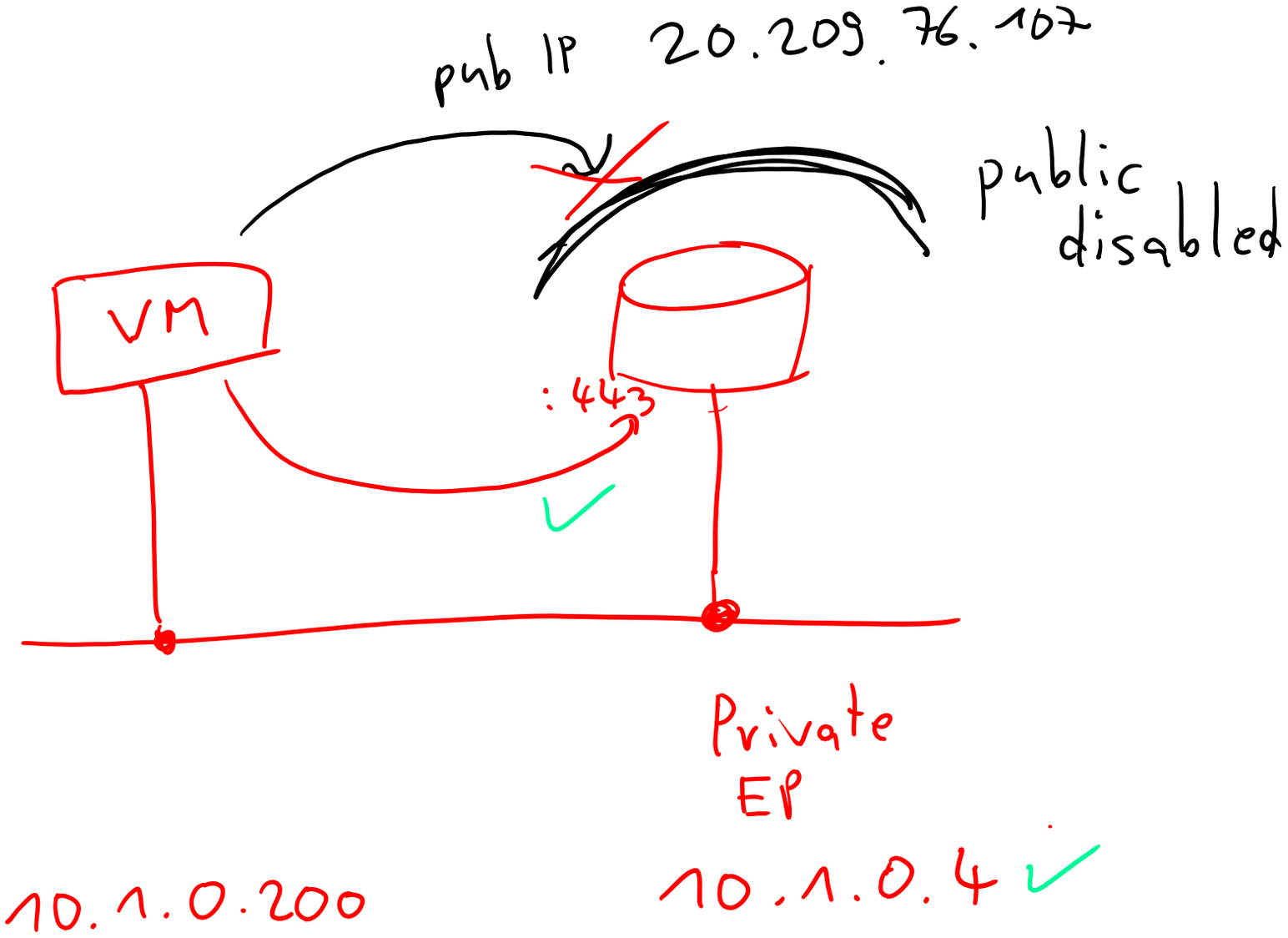

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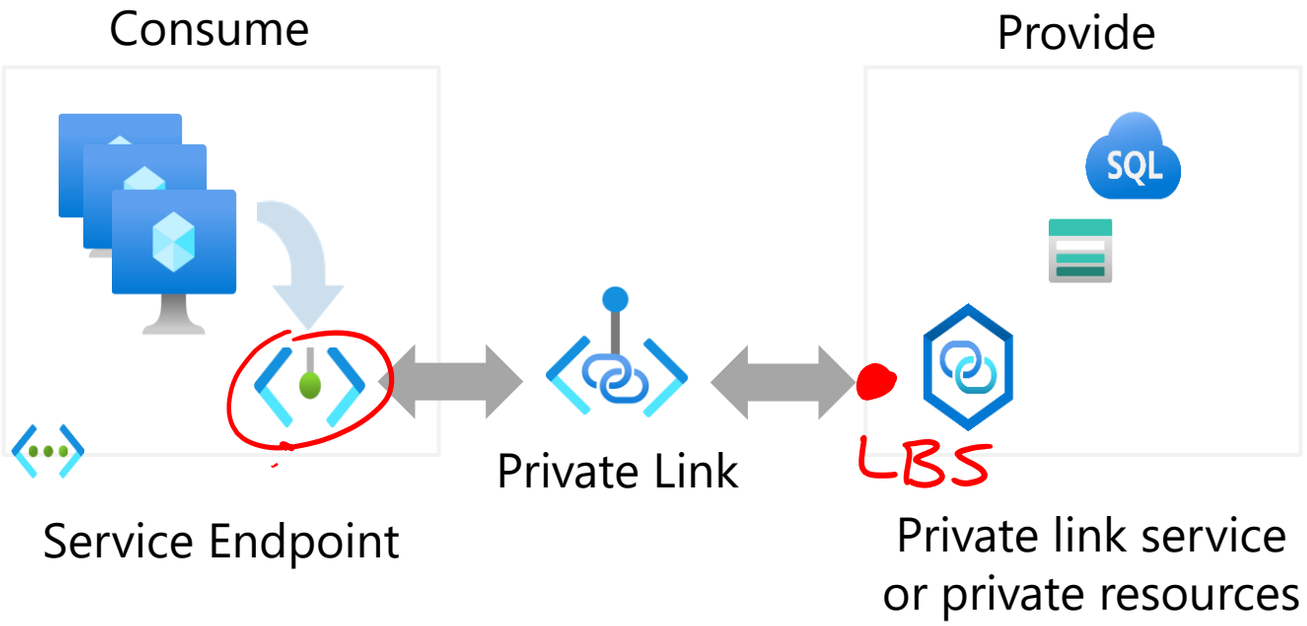
What is Azure Private Link?

- Build and allow private connections between resources
- To privately connect to a service, create a service endpoint
- To privately render a service, create a private link service or private resources
- All data travels on the Microsoft backend



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Compare Service Endpoints and Private Endpoints

Feature	Service Endpoint	Private Endpoint
Use Case	Simple, cost-effective way to secure traffic to Azure services from within VNet	Maximum security, regulatory compliance, hybrid connectivity (on-prem to private IP).
Security	Secures traffic from VNet to service, but the service endpoint remains publicly accessible (can be firewall-restricted).	Highest security; traffic stays within VNet/Microsoft backbone, no public exposure; excellent for data exfiltration prevention.
On-premises access	No direct access (via public IP)	Yes (via VPN/ExpressRoute to private IP)
IP Address	Public IP of service (traffic uses private VNet IP to reach it)	Private IP from your VNet
DNS	No change needed (resolves to public IP)	Requires Private DNS Zone
Cost	Free (no added cost)	Cost for endpoint & data processed

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Lab - Implement Intersite Connectivity

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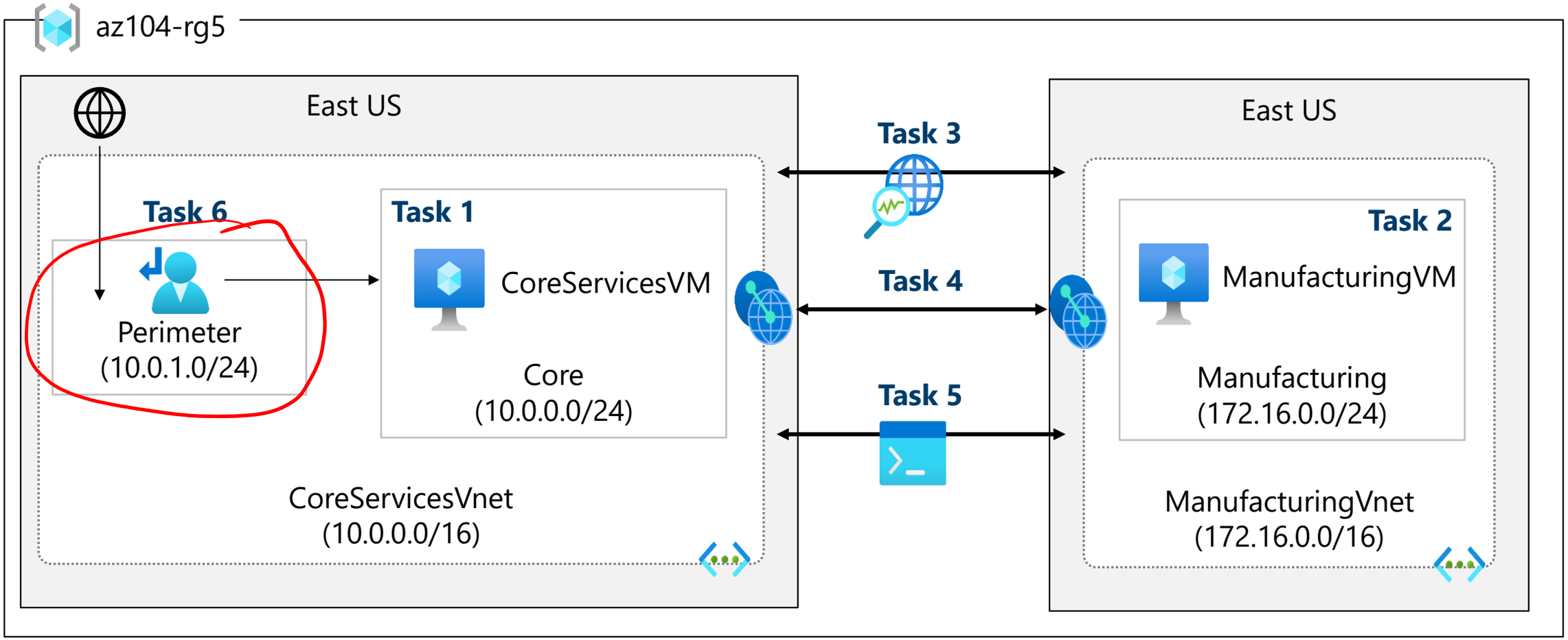
Lab 05 – Architecture diagram



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